

Re-visiting Brenier's Theorem

Brenier's Theorem

Assumptions:

1. $c(x, y) = \|x - y\|^2$
2. α has a density $\rho_\alpha = \frac{d\alpha}{dx}$



Monge's Problem:

(Continuous / general version)

$$\min_{\beta = T_\# \alpha} \int_X c(x, T(x)) d\alpha(x)$$

Theorem [Brenier, 1991]:

1. There exists a **unique** map T that solves the Monge problem
2. T is the gradient of a convex function

Formally: the unique $T = \nabla \varphi$ such that φ is **convex** and $(\nabla \varphi)_\# \alpha = \beta$

